

F1 RACE PREDICTOR

Machine Learning Analytics Learning Guide

Understanding the 5 Statistical Visualizations

GRAPH 1: FEATURE IMPORTANCE (Bar Chart)

This bar chart ranks features by their predictive power in the ML model. Driver average points (18.34%) and grid position (15.67%) emerge as top features, showing that historical driver performance and qualifying results are the strongest indicators for predicting race winners.

GRAPH 2: DISTRIBUTION ANALYSIS (Three Histograms)

These three histograms visualize how different features are distributed across the dataset. Driver performance shows right-skewed distribution (concentrated at lower values), grid position displays near-uniform spread, and constructor performance follows normal distribution—revealing that each feature has distinct statistical properties valuable for model training.

GRAPH 3: CLASS DISTRIBUTION (Three Pie Charts)

These pie charts expose class imbalance in prediction tasks: only 4.2% of races end in a specific winner outcome, 12.7% result in podium finishes, and 42.1% achieve top-10 placement. This imbalance requires stratified sampling techniques to prevent the model from becoming biased toward the majority (non-winner) class.

GRAPH 4: CORRELATION ANALYSIS (Two Scatter Plots)

The first scatter plot shows grid position vs. final position with a Pearson correlation of 0.951—an exceptionally strong linear relationship meaning qualifying results are highly predictive of race outcomes. The second plot reveals logarithmic correlation between feature importance scores and model accuracy, validating that our feature engineering methodology is sound.

GRAPH 5: FEATURE IMPORTANCE HEATMAP (Task-Specific Ranking)

This heatmap compares feature importance across three prediction tasks (winner, podium, top-10). Driver average points consistently ranks highest (18.34% for winner, 10.45% for podium, 22.34% for top-10), demonstrating that this feature's predictive power holds true across different classification boundaries—a key insight that validates our model's generalization ability.

KEY INSIGHTS FROM THE VISUALIZATIONS

Feature Engineering Works: Top 15 features capture ~90% of predictive power, confirming careful feature selection was effective.

Historical Performance Matters Most: Driver average points and past performance are stronger predictors than qualifying position alone.

Class Imbalance is Significant: Only 4.2% winners requires stratified cross-validation and balanced accuracy metrics.

Strong Linear Relationships Exist: Grid-to-finishing position correlation (0.951) indicates qualifying directly influences outcomes.

Task-Specific Patterns Emerge: Feature importance varies across prediction tasks, suggesting different mechanisms drive each outcome type.